

When Sitting Still Leads to a Standstill: The Role of Physical Activity and Sedentary Time in Coping with Academic Stress

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University students face considerable academic and psychosocial stressors, placing them at elevated risk for mental health challenges compared to non-student peers (Sheldon et al., 2021). Physical activity (PA) has been shown to buffer stress and promote well-being in students, whereas sedentary time (ST) is linked to adverse health outcomes (Wunsch et al., 2021). However, little is known about how PA and ST relate to patterns of academic behavior and experience reflecting students' coping with academic demands. This study investigated the effects of PA and ST on transitions between academic behavior and experience patterns over time.

We used data from the Lübeck University Student Trial (LUST), a longitudinal cohort study monitoring student health annually. The sample comprised 1,965 observations from 785 students ($M_{\text{age}} \pm \text{SD}$: 20.81 ± 3.33) who participated at least twice between 2022 and 2025. Academic behavior and experience patterns were assessed using a student-adapted version of the Work-related Coping Behavior and Experience Patterns questionnaire (Schaarschmidt & Fischer, 1997), classifying respondents into health-promoting (G: Good health, S: Sparing investment) or health-endangering coping patterns (A: Ambitious, B: Burnout). PA and ST were measured using the International Physical Activity Questionnaire short form (Craig et al., 2003) and operationalized as weekly metabolic equivalent of task minutes and average daily sitting hours, respectively. Continuous-time multi-state Markov models were applied to estimate the effects of PA and ST on transition probabilities between patterns over time, modeling all patterns as transient states and allowing transitions between any pair of states.

Higher ST was associated with an increased hazard of transitioning from pattern G to B within one year (HR = 2.30, 95% CI: 1.15 – 4.59). The probability of B being the subsequent pattern following G was 3.5% at low ST (-1 SD) and 16% at high ST (+1 SD). PA was not associated with transition hazards between patterns. The findings necessitate further validation due to partially limited transition event numbers.

Reducing structurally mandated sitting may support more stable and health-promoting coping patterns in academic settings. Universities could benefit from implementing active campus policies to strengthen students' coping capacity and prevent progression toward burnout-related coping.